

Curvewise Conformal Prediction Bands for Periodic PMSM Torque Surrogates under Design-Distribution Shift

Online Resource 1 for journal-neutral review - anonymous review copy

The released archive (DOI 10.5281/zenodo.15688397) contains 23,250 independent simulated geometries from one two-dimensional, quarter-symmetry PMSM model. Each geometry has 20 inputs and a 120-angle torque period. The archive is licensed GPL-3.0-or-later. These are simulator rows, not independent machines or hardware measurements.

Published table	Study role	Designs	Torque range	Median retained Fourier energy
train_test	development/internal	2,000	0.297922-0.695034	99.999955%
uq_uniform	large uniform evaluation	11,250	0.296617-0.702839	99.999955%
uq_gauss	Gaussian-shift evaluation	10,000	0.340070-0.681911	99.999920%

All values were finite. The audit found no duplicate parameter rows, no duplicate torque rows, and zero exact parameter-row overlap in each of the three pairwise table comparisons. All files shared the 0-29.75 degree grid at 0.25-degree increments. The audit establishes row integrity and absence of exact reuse; it does not establish fidelity beyond the source simulator.

Dataset	Role	File	Bytes	SHA-256
train_test	parameters	train_test_parameters.csv	477,742	10c0b1b344d5addadacfad2b53be7d41e70f4e6ecc9fa8344fd840fbe8906a0
train_test	torque	train_test_torque.csv	2,856,040	f9606c7e6b386e5eaf6537d63a745973d31fee6cb907101b85cd2c2d864cf39f
uq_uniform	parameters	uq_uniform_parameters_11k.csv	2,686,443	020f4b85fff4475f9662b1246c85c3253398fbef8c2b72b4f59d69b8fa262e88
uq_uniform	torque	uq_uniform_torque_11k.csv	16,061,335	df1f8aa5bf6e139659a7f935fb31b3831c2403119086ebaa323af83d3242656c
uq_gauss	parameters	uq_gauss_parameters_10k.csv	2,387,927	368832b4f571385788320c864a8a7a425da818b1acb5e66b6726e0d529088a40
uq_gauss	torque	uq_gauss_torque_10k.csv	14,277,285	ff85a001286dda41455e08d7ed6b269d7f0c9f16184f701af141cf848e3098e7

Rows 1-1,800 of `train_test` form the development pool. Seed 20260821 selects 1,200 fit rows and 600 calibration rows; the final 200 rows remain an internal evaluation set. The 11,250 uniform and 10,000 Gaussian rows never select a model or hyperparameter. Four additional deterministic seeds (1201, 2402, 3603, and 4804) repeat only the fit/calibration split.

The primary endpoint is simultaneous full-curve coverage at nominal 90% ($\alpha=0.10$); 95% is a frozen sensitivity analysis. A design is covered only when all 120 torque angles fall inside its band. The nonconformity score is the maximum absolute error across the period. Wilson intervals treat independently simulated design rows as binomial units; they do not represent physical-machine variation.

The geometry scale uses the fifth-nearest fit-design distance after range normalization, divided by the median fit leave-one-out fifth-neighbour distance and floored at 0.25. The support warning compares an evaluation distance with the 600 calibration distances and flags an upper-tail p-value below 0.05. Neither diagnostic uses evaluation torque labels.

All methods predict the same 21 real coordinates corresponding to 11 retained DFT components and reconstruct all 120 angles. Candidate settings for ridge, kernel ridge, and Extra Trees were selected by three-fold waveform-MAE cross-validation using only the 1,200 fit designs. The ARD Gaussian process optimized a shared 20-dimensional RBF kernel by fit-only log marginal likelihood with length-scale bounds 0.03-10, numerical noise $1e-8$, and no restarts. It is source-style, not an exact reproduction of the source article's coordinate-wise GP/CMA-ES workflow.

Model	Selected parameters	Selection basis
poly2_ridge	{'alpha': 1e-07}	three-fold fit-only waveform MAE
rbf_kernel_ridge	{'alpha': 0.0001, 'gamma': 0.025}	three-fold fit-only waveform MAE
ard_gaussian_process	{'dimensions': 20.0, 'length_scale_upper': 10.0}	fit-only log marginal likelihood
extra_trees	{'n_estimators': 300.0, 'min_samples_leaf': 1.0, 'max_features': 1.0}	three-fold fit-only waveform MAE

The table reports both frozen nominal levels. Widths are half-widths in the units stored by the source archive. Accepted-set coverage is descriptive because support screening changes the evaluated population.

Evaluation	Band	Nominal	n	Full-curve coverage	95% Wilson interval	Mean half-width	Support rejected	Accepted-set coverage
fixed internal uniform	global	95%	200	96.00%	92.31%-97.96%	0.009615	5.000%	97.37%
fixed internal uniform	geometry-scaled	95%	200	96.00%	92.31%-97.96%	0.009028	5.000%	97.37%

Evaluation	Band	Nominal	n	Full-curve coverage	95% Wilson interval	Mean half-width	Support rejected	Accepted-set coverage
fixed internal uniform	global	90%	200	92.00%	87.40%-95.02%	0.007125	5.000%	93.68%
fixed internal uniform	geometry-scaled	90%	200	92.00%	87.40%-95.02%	0.006834	5.000%	93.68%
large uniform	global	95%	11,250	95.51%	95.11%-95.88%	0.009615	5.511%	96.50%
large uniform	geometry-scaled	95%	11,250	95.25%	94.84%-95.63%	0.008991	5.511%	96.06%
large uniform	global	90%	11,250	90.29%	89.73%-90.83%	0.007125	5.511%	91.85%
large uniform	geometry-scaled	90%	11,250	89.96%	89.40%-90.51%	0.006806	5.511%	91.23%
Gaussian density shift	global	95%	10,000	100.00%	99.96%-100.00%	0.009615	0.000%	100.00%
Gaussian density shift	geometry-scaled	95%	10,000	100.00%	99.96%-100.00%	0.007400	0.000%	100.00%
Gaussian density shift	global	90%	10,000	100.00%	99.96%-100.00%	0.007125	0.000%	100.00%
Gaussian density shift	geometry-scaled	90%	10,000	100.00%	99.96%-100.00%	0.005601	0.000%	100.00%

Each uniform quintile contains 2,250 independent simulations and each Gaussian quintile 2,000. Geometry-distance quintiles are descriptive conditional audits, not additional calibration groups. The primary failure is visible in uniform quintile 5: global coverage is 76.13% and scaled coverage 78.31%, despite near-nominal aggregate coverage.

Evaluation	Distance quintile	n	Mean max error	Global coverage	Scaled coverage	Mean scaled half-width
large uniform	1	2,250	0.002693	97.96%	96.84%	0.006207
large uniform	2	2,250	0.003181	95.47%	94.13%	0.006557
large uniform	3	2,250	0.003620	92.58%	91.42%	0.006790
large uniform	4	2,250	0.004109	89.33%	89.11%	0.007035
large uniform	5	2,250	0.005522	76.13%	78.31%	0.007438
Gaussian density shift	1	2,000	0.002281	100.00%	100.00%	0.005414
Gaussian density shift	2	2,000	0.002344	100.00%	100.00%	0.005534
Gaussian density shift	3	2,000	0.002422	100.00%	100.00%	0.005604
Gaussian density shift	4	2,000	0.002477	100.00%	100.00%	0.005671
Gaussian density shift	5	2,000	0.002509	100.00%	100.00%	0.005781

No seed was selected from target performance. The table retains all 30 primary ARD rows (five seeds by three evaluations by two bands). Uniform global coverage ranges from 87.99% to 91.08%; scaled coverage ranges from 88.93% to 91.31%. The sparse-tail limitation remains because every split samples the same simulator and released design mechanism.

Seed	Evaluation	Band	Coverage	Mean half-width
20260821	fixed internal uniform	global	92.00%	0.007125
20260821	fixed internal uniform	geometry-scaled	92.00%	0.006834
20260821	large uniform	global	90.29%	0.007125
20260821	large uniform	geometry-scaled	89.96%	0.006806
20260821	Gaussian density shift	global	100.00%	0.007125
20260821	Gaussian density shift	geometry-scaled	100.00%	0.005601
1201	fixed internal uniform	global	89.50%	0.006826
1201	fixed internal uniform	geometry-scaled	89.50%	0.006757
1201	large uniform	global	88.83%	0.006826
1201	large uniform	geometry-scaled	88.93%	0.006708
1201	Gaussian density shift	global	100.00%	0.006826
1201	Gaussian density shift	geometry-scaled	99.96%	0.005420
2402	fixed internal uniform	global	88.00%	0.007257
2402	fixed internal uniform	geometry-scaled	87.50%	0.007051
2402	large uniform	global	91.08%	0.007257
2402	large uniform	geometry-scaled	91.04%	0.007025
2402	Gaussian density shift	global	100.00%	0.007257

Seed	Evaluation	Band	Coverage	Mean half-width
2402	Gaussian density shift	geometry-scaled	100.00%	0.005743
3603	fixed internal uniform	global	88.00%	0.007550
3603	fixed internal uniform	geometry-scaled	89.00%	0.007680
3603	large uniform	global	90.41%	0.007550
3603	large uniform	geometry-scaled	91.31%	0.007624
3603	Gaussian density shift	global	100.00%	0.007550
3603	Gaussian density shift	geometry-scaled	99.98%	0.006214
4804	fixed internal uniform	global	87.50%	0.006446
4804	fixed internal uniform	geometry-scaled	87.50%	0.006650
4804	large uniform	global	87.99%	0.006446
4804	large uniform	geometry-scaled	89.23%	0.006600
4804	Gaussian density shift	global	99.99%	0.006446
4804	Gaussian density shift	geometry-scaled	99.95%	0.005379

This analysis was added after inspecting the primary aggregate results and uses the degree-two ridge surrogate. Each target table is split in half: one half fits a quadratic logistic domain classifier from geometry alone, and the other evaluates bands. The test weight is retained as point mass at positive infinity; an infinite quantile is not clipped. Coverage 'including vacuous' is set-theoretic and has no engineering utility when the finite-band rate is zero.

Target	Method	Nominal	n	Finite-band rate	Coverage incl. vacuous	Finite-band coverage	Mean finite half-width
large uniform	global	90%	5,625	100.00%	90.40%	90.40%	0.023903
large uniform	geometry-scaled	90%	5,625	100.00%	91.09%	91.09%	0.024129
large uniform	estimated weighted	90%	5,625	100.00%	91.98%	91.98%	0.025025
large uniform	global	95%	5,625	100.00%	95.72%	95.72%	0.030677
large uniform	geometry-scaled	95%	5,625	100.00%	95.70%	95.70%	0.029914
large uniform	estimated weighted	95%	5,625	100.00%	95.77%	95.77%	0.030904
Gaussian density shift	global	90%	5,000	100.00%	96.68%	96.68%	0.023903
Gaussian density shift	geometry-scaled	90%	5,000	100.00%	90.88%	90.88%	0.019873
Gaussian density shift	estimated weighted	90%	5,000	0.00%	100.00%	not defined	not defined
Gaussian density shift	global	95%	5,000	100.00%	99.60%	99.60%	0.030677
Gaussian density shift	geometry-scaled	95%	5,000	100.00%	97.10%	97.10%	0.024637
Gaussian density shift	estimated weighted	95%	5,000	0.00%	100.00%	not defined	not defined

Target	Weight role	Designs	ESS	Maximum normalized mass	Held-out domain AUC
large uniform	source_calibration	600	516.50	0.529%	0.5059
large uniform	target_evaluation	5,625	4828.08	0.092%	0.5059
Gaussian density shift	source_calibration	600	4.31	39.704%	1.0000
Gaussian density shift	target_evaluation	5,000	2741.88	0.171%	1.0000

For the uniform half-split, domain AUC was 0.5059, calibration ESS 516.50/600, and all 5,625 primary 90% weighted bands were finite. For the Gaussian half-split, domain AUC was 1.0000, calibration ESS 4.31/600, one source row carried 39.70% of normalized mass, and 0/5,000 weighted bands were finite at either nominal level. This is retained as an overlap failure, not repaired.

Timings are workstation-specific and exclude high-fidelity simulation. The ARD model's 41.0 s figure includes fitting and fit-only hyperparameter optimization; predicting 11,250 uniform designs required 0.317 s.

Model	Evaluation	Fit/select seconds	Predict seconds	Designs
poly2_ridge	fixed internal uniform	0.110	0.001	200
poly2_ridge	large uniform	0.110	0.028	11,250
poly2_ridge	Gaussian density shift	0.110	0.026	10,000

Model	Evaluation	Fit/select seconds	Predict seconds	Designs
rbf_kernel_ridge	fixed internal uniform	0.990	0.005	200
rbf_kernel_ridge	large uniform	0.990	0.177	11,250
rbf_kernel_ridge	Gaussian density shift	0.990	0.147	10,000
ard_gaussian_process	fixed internal uniform	41.005	0.008	200
ard_gaussian_process	large uniform	41.005	0.317	11,250
ard_gaussian_process	Gaussian density shift	41.005	0.260	10,000
extra_trees	fixed internal uniform	3.178	0.069	200
extra_trees	large uniform	3.178	0.164	11,250
extra_trees	Gaussian density shift	3.178	0.140	10,000

Representative curves were selected mechanically by frozen role and median-error rules rather than by visual preference.

Distribution	Role	Published row	Max error	Global width	Scaled width	Global covered	Scaled covered
uq_uniform	dense_both_covered	607	0.002257	0.007125	0.006385	1	1
uq_uniform	sparse_scaled_rescue	7690	0.007269	0.007125	0.008054	0	1
uq_uniform	sparse_both_missed	665	0.010097	0.007125	0.007273	0	0
uq_gauss	gaussian_middle_both_covered	6106	0.002470	0.007125	0.005626	1	1

The companion bundle includes the analysis modules, deterministic scripts, tests, frozen aggregate and per-design results, figures, manuscript sources, and SHA-256 manifest. Raw public data are omitted from the bundle to avoid redistribution and must be downloaded from the source DOI. The main evidence validator recomputes claims from 42,900 primary per-design rows and the disjoint weighted diagnostic.

The evidential scope is one public two-dimensional PMSM simulator. The study does not quantify simulator model-form discrepancy, three-dimensional effects, manufacturing variation, another topology, another machine, or hardware uncertainty. Several ARD length scales reached the optimization upper bound. The weighted ratio is estimated, not known, and the diagnostic is explicitly post-primary.

Frozen evidence file	SHA-256
papers/paper2_torque_uq/manuscript.md	01699a7a7010f239bf900c51fa7efc02b3a4501e5e2ff47d31955c1ef8e8f6ab
results/paper2_torque_conformal/aggregate_summary.csv	0700590d03103d3cc7b68b55a2d89afa5f244e08f85cee41950ccf5f175ccd7
results/paper2_torque_conformal/primary_per_design_scores.csv.gz	a5695affa0244d2d4586db095ec2277ec1aec8b7533fa3b25d03e7833fe5f905
results/paper2_torque_seed_sensitivity/aggregate_summary.csv	424b57d378301f325ae940ec2013dce60368f04cc5c95a4b367caa8afdee665c
results/paper2_weighted_conformal/comparison_summary.csv	a5f08fe15d3e6dfbe3cd49517f6b04033b6a91400ff7243beb0410cb4a4fb3c7
results/paper2_weighted_conformal/weighted_per_design.csv.gz	e3974ca6697b4e2b6fd283e39730546687bf63ba102a18869133272679d06da0
results/paper2_evidence_validation/evidence.json	deb721c39280d53fc4c302ecaebca72fba913e724da8720ba1ff7191975a8dbc